Module - III

Expert System in AI

Expert System

An expert system is a computer program that uses AI techniques to simulate the decision making ability of a human expert.

It is design to solve complex problems in a specific domain by reasoning through bodies of knowledge, mainly represented as if-then rules- rather than conventional procedural code.

In short,

Human Expert ⟶ Knowledge.

Expert System ⟶ Software Version of the Expert.

Components of Expert System:

1. Knowledge Base
2. Inference Engine
3. User Interface

1. Knowledge Base

Stores fact and rules about the specific domain. For example: if patient has fever and cough then possible flu.

2. Inference Engine

The reasoning mechanism of expert system.

Applies logical rules to the knowledge base to deduce new facts or reach conclusions.

For example: uses forward chaining and backward chaining to find solutions.

3. User Interface

The medium through which the user interacts with the expert system.

Provides explanation and takes user queries.

Characteristics of Expert System:

1. High performance in narrow domain.
2. Provides explanation for its decisions.
3. Handles uncertain or incomplete information.
4. Knowledge driven not just data driven.

Types of Expert System:

Expert System can be classified into different types.

1. **Rule Based Expert System**

   Knowledge represented as rules (if-then).

   Example: MYCIN (Medical Diagnosis).

2. **Frame Based Expert System**

   Used objects and attributes (frame) instead of rules.

   Example: XCON for configuring computer systems.

3. Fuzzy Expert System

Deals with uncertainity and imprecise information using fuzzy logic.

Example: Climate control systems.

4. Model Based Expert System

Relies on a model of the physical world to reason.

Example: System diagnosing faults in electric circuit.

5. Hybrid Expert System

Combines rule-based, neural networks and fuzzy logics.

Example: Medical expert systems integrating multiple reasoning approaches.

Advantages of Expert System:

3. Fuzzy Expert System

Deals with uncertainity and imprecise information using fuzzy logic.

Example: Climate control systems.

4. Model Based Expert System

Relies on a model of the physical world to reason.

Example: System diagnosing faults in electric circuit.

5. Hybrid Expert System

Combines rule-based, neural networks and fuzzy logics.

Example: Medical expert systems integrating multiple reasoning approaches.

Advantages of Expert System:

1. Consistency

Provides the same solution to the same problem, unlike humans who may vary.

2. Availability

Can be used anytime 24X7.

3. Speed & Efficiency

Processes large knowledge quickly.

4. Decision Support

Helps expert in-making better decision.

5. Preserves Knowledge

Knowledge of experts can be stored and reused.

6. Training & Education

Used as teaching tools.

Disadvantages of Expert System:

1. Lack of Creativity

Cann't think beyond the programmed knowledge.

2. High Development Cost

Building and maintaining is expensive.

3. Limited Domain

Works well only in specific, narrow fields.

4. No Common Sense

Unlike humans, it can not adapt easily to new situations.

Applications of Expert System:

1. **Medicine**

   Diagnosis, drug prescription.

2. **Engineering**

   Fault diagnosis in machinery.

3. **Business & Finance**

   Credit approval, investment advice.

4. **Agriculture**

   Crop disease identification, soil management.

5. **Weather Forecasting**

   Predicting climate using fuzzy expert system.

Expert Knowledge Acquisition in AI

Knowledge Acquisition in AI is the process of extracting, organizing and formalizing knowledge from human experts, documents, databases, or the real world into a form that a machine can use.

Its about capturing the expertise of a human specialist and putting it into an AI/expert system, so that the machines can solve problems like an expert.

Example: Taking doctors diagnostics routes and encoding them into a medical expert system.

Importance:

1. An expert system is only as good as its knowledge base.
2. Without proper knowledge acquisition, the system may be incomplete, inconsistent or wrong.
3. The process solves the "knowledge acquisition bottle-neck", a big challenge in AI.

Process of Knowledge Acquisition:

1. **Identify Domain & Experts**

   Choose a field (medicine, engineering, finance, etc) and consult domain experts.

2. **Elicit Knowledge**

   Extract experts reasoning, heuristic and decision rules.

3. **Formalize Knowledge**

   Convert into rules and frames.

4. **Validate Knowledge**

   Test system output against expert judgement.

5. **Refine & Update**

   Improve over time as knowledge evolves.

Types of Knowledge Acquisition:

1. Manual (Human Centered) Method

(i) **Interviews**: Structured / Unstructured question & answer with expert.

(ii) **Observation**: Watching experts solving real problems.

(iii) **Protocol Analysis**: Asking experts to "Think Aloud" while solving a problem.

(iv) **Questionaries / Surveys**: Collecting structured responses.

2. Automated (Machine Centered) Method

(i) **Machine Learning**: Extracting knowledge from large datasets automatically.

(ii) **Data Mining**: Discovering patterns and rules from databases.

(iii) **Natural Language Processing**: Extracting knowledge from text or documents.

3. Hybrid Method

Combines human expert input with machine learning to refine rules. For example, doctors provide rules and ML refines them with patient data.

Types of Knowledge Captured:

1. **Factual Knowledge**

   Facts, Truths. Eg: Water boils at 100°C.

2. **Heuristic Knowledge**

   Experienced-based, Rule of thumb strategies.
   Example: If chest pain & shortness of breath, possible hearth issue.

3. **Procedural Knowledge**

   Step by step process.

4. **Meta Knowledge**

   Knowledge about how to use Knowledge.
   Eg: Which rule applies best.

Expert System in Mechanical Engineering Applications

Introduction:

An Expert System in AI is a computer program that simulates human expert's problem solving ability using knowledge-based reasoning.

In mechanical engineering, expert systems are used to support design, manufacturing, planning, automation and control.

They help engineers make quick, consistent and optimized decisions in complex-technical environment.

Example: An expert system in CAD (Computer Aided Design) can suggest the best material and design modifications for weight reduction while maintaining strength.

Applications:

1. **CAD (Computer Aided Design)**

   Role of Expert System:

   (i) Suggests design alternatives.
   (ii) Checks feasibility, safety and manufacturing.
   (iii) Automates repetitive design tasks.

   Example:

   - Knowledge-based CAD tools that automatically design gears, bearings or heat exchangers.

2. CAPP (Computer Aided Process Planning)

Role of Expert System:

(i) Generate process plans automatically from CAD data.

(ii) Chooses machine sequence, tools and cutting parameters.

(iii) Optimizes cost and time.

Example:

An expert system in machining suggest cutting speed, feed rate and tool material for CNC machining of titanium alloys.